Enhanced sales forecasting of perishable and non- perishable goods using weather data with insights on Imports and price sensitivity

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Abstract: **In the competitive restaurant industry, success hinges on accurate demand forecasting and optimal pricing strategies. Weather factors such as temperature, rain, and seasonal changes significantly impact consumer behavior, particularly affecting sales of goods categorized as highly perishable (e.g., fresh produce, dairy), perishable (e.g., bread, meat), and non-perishable (e.g., canned goods, dry ingredients). The project leverages historical weather patterns, ingredient prices, and machine learning to forecast sales and implement dynamic pricing, with a focus on import status and price sensitivity. Using Random Forest and ARIMA models, the system conducts time series analysis for precise demand prediction. Built with Python and utilizing libraries like scikit-learn and pandas, the project processes real-time weather and ingredient price data to enable timely price adjustments and inventory management. The data-driven approach empowers restaurants to maximize revenue, optimize menu pricing, and enhance customer satisfaction in a fluctuating market.**

Keywords --- Demand forecasting, dynamic pricing, weather data, machine learning, time series analysis, Random Forest, ARIMA, highly perishable goods, perishable goods, non-perishable goods, import status, price sensitivity, restaurant sales, real-time data.

I. INTRODUCTION

This fast-growing restaurant environment is forcing the operations to be always better in knowing the customer and predicting his behavior and decision-making. Two forces which were minor factors relatively, weather and ingredient prices, have now become major contributors to restaurant sales. Weather can decide when and how frequently customers go out to dine, for favorable conditions boost dining out, while unfavorable weather hinders it. Meanwhile, fluctuations in the price of ingredients pressure the margins, and thus strategic changes have to be made in order to stay profitable as well as deliver customer satisfaction.

Of major challenge, restaurant industry is faced with optimizing sales of perishable and non-perishable as well as import goods due to dynamic factors like weather, demand patterns, and price sensitivity. Traditional models cannot give accurate demand forecasting; instead, product overstocking or stockouts, which is a significant problem for perishable products with low shelf lives, occurred. This project will erase all those problems by using machine learning algorithms like Time Series Forecasting and Dynamic Pricing in predicting the amounts of demand based on local weather forecast data and types of product (perishable, non-perishable, and imported goods), ingredient costs, and other exogenous factors. The model returns an MSE value of [53.97], which yet again confirms that the model is highly accurate.

Most conventional sales forecasting models do not account for real-time situational variables, such as weather and ingredient price fluctuations. This approach integrated various sources of data including, but not limited to historical sales, weather conditions, and fluctuating prices, thereby generating a more holistic model. The model shows how the external drivers are influencing customer demand over time with accurate amalgamation of temperature, humidity, and rainfall data integrated with trends of price that could be influenced by market movements, supply chain disruptions, or seasonality. For instance, change in the weather can influence the demand for seasonal dishes, such as hot soups during winter or light meals in summer. This is something restaurants can plan their seasonal menus or promotions around.

It develops a dynamic pricing model that should change the prices of foods on the menu according to fluctuating weather conditions and ingredient costs in real time, helping restaurants maintain profit margins even as they stay competitive-the tactic, more commonly employed in such industries as airlines and hotels, has been relatively new to restaurants. This model helps restaurants make smooth responses to changes in demand, ingredient costs, and other factors going into determining their business; therefore, profit margins will not be compromised at the cost of customer satisfaction.

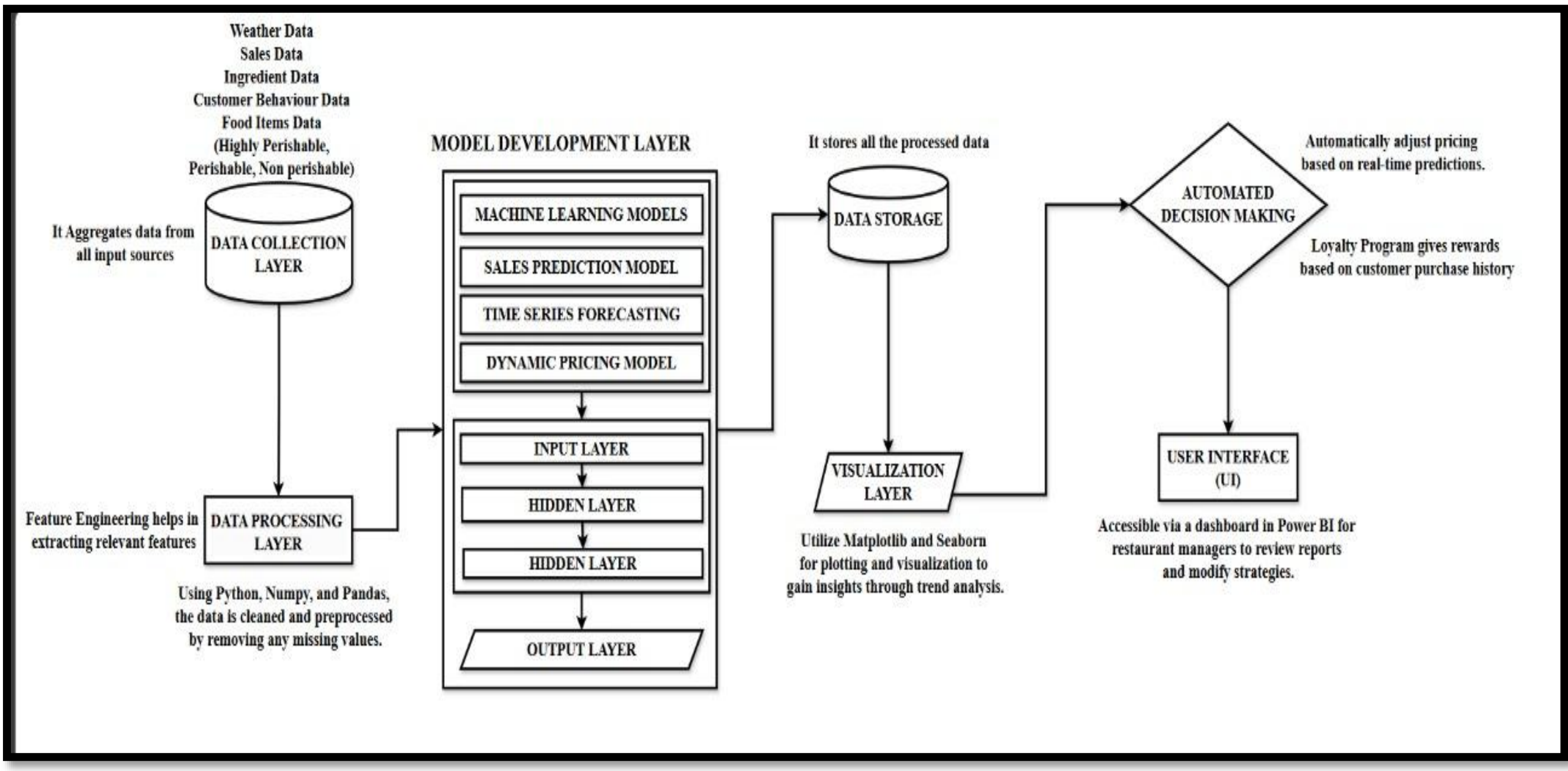

Figure 1: System Architecture

Additionally, it can depict an overall view of whether weather conditions or price sensitivity affects sales so that informed decisions based on data can be made. Unlike its competitors that do not change when abrupt weather conditions or special events happen, the project will permit changes on real-time basis in order to minimize waste and maximize sales. This advantage will ensure accuracy in giving information to optimize everything in managing inventory, reduce costs, and assure satisfaction of customers' needs. The dynamic pricing strategies, real-time data analysis, and integration of machine learning methods would thus make the restaurant business highly adaptable and successful in this volatile market.

The system will also facilitate the implementation of customer engagement strategies. For example, a restaurant can offer special deals for hot drinks when it is cold or cool beverages when there is a heatwave. Targeted promotions increase the level of customer involvement and their loyalty, which increases repeat visits and cash flow generation.

For these insights to be actionable, the project applies interactive dashboards with products such as Power BI. The interactive dashboards allow restaurant managers to understand trends and behavior patterns of customers so that they can make proper pricing decisions in attaining optimal operational efficiency. It gives restaurants an integrated decision-support system in consolidated models of weather conditions, ingredient prices, and sales data for proper sales predictions, cost management, and customer relationships.

This model therefore addresses the weaknesses of the traditional sales forecasting since multiple external factors are all incorporated into one dynamic framework.

The advantage over inventory and cost management lies in its dynamic pricing and real-time adjustments, so restaurants are confident about long-term success and continued profitability.

The architecture diagram of the system is illustrated in Figure 1. The system has multiple layers aimed at the integration of data collection, model development, and decision-making processes:

1. Data Collection : Various sources of data, such as weather conditions, sales data, ingredients data, and price sensitivity are combined. The data is categorized into highly perishable, perishable, and non-perishable goods, thus offering insights into product demand patterns.

2. Data Processing Layer : This layer of data processing is aimed at making the processed data ready for analysis. For preprocessing purposes, this layer relies on relevant Python libraries like Pandas and Numpy and the tasks can include data cleansing, data imputation, and feature engineering among others aimed at extracting informative features for modeling.

3. Model Development Layer : This core part includes models that would be based on learning machines for sales prediction, time series forecasting, and dynamic pricing. It's always trained on historical data, so the system can predict future trends in sales and change its prices accordingly.

4. Data Storage: This section will store the processed data for any further analysis needed for the next step.

5. Visualization : This layer uses Matplotlib and Seaborn to form visual insights such as trend analysis.

6. Automated Decision Making: The real-time prediction-based system can automatically adjust prices as well as provide loyalty programs that award customers according to their history of purchases, so more customers will get engaged.

7. User Interface (UI): Power BI demonstrates the processed information to the user on the dashboard. Restaurant managers can then track reports and change strategies according to new insights for that report.

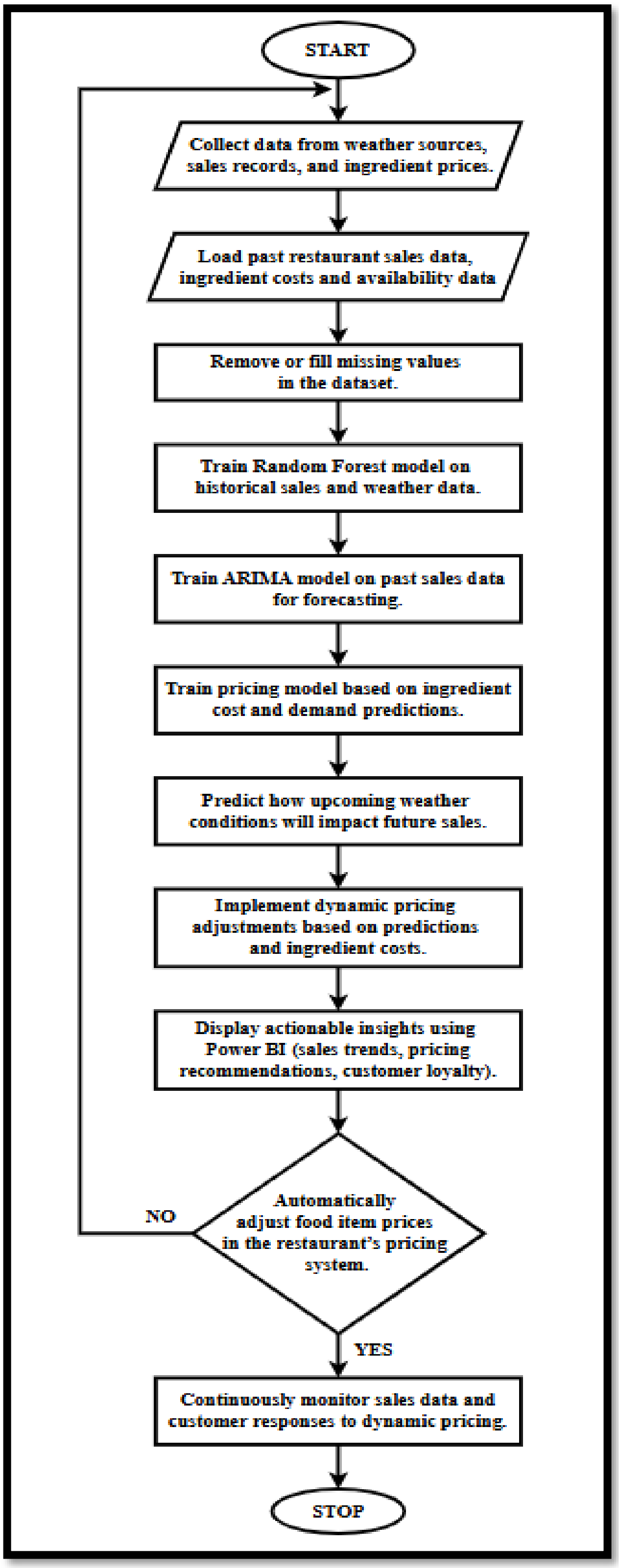

Figure 2:Workflow

Figure 2 Overall Process Using a Workflow How It Processes All Data, Including Collection up to the Adjustment of Dynamic Pricing actual implementation:

1. Source Data Collection: Weather data, prior sales history, ingredient cost data, and even availability would be collected.
2. Upload Previous Data: Upload prior sales data as well as other relevant data to be used for analysis
3. Data Preprocessing: Cleaning up the data, proper treatment of missing entries, and other such which shall make the data of better quality for model training.
4. Train a sequence of Random Forest model in predicting the sales based on historical data and then follow it by ARIMA to predict future values of time-series.
5. Dynamic Pricing Model Train pricing model on the predictions made by sales and the cost of the ingredients involved.
6. Weather Impact Prediction Generate the forecast as to how upcoming changes in the weather will influence sales.
7. Dynamic Price Adjustment The prices adjust automatically depending on the prediction of demand and the expected cost.
8. Power BI Dashboards: Visualizing actionable insights on sales trend, pricing strategy, and customer loyalty for decision making.
9. Automatic Price Updates: Real-time adjustment of food item price in the restaurant's price list.
10. Continuous Adaptation: Continuously monitor incoming sales data and customer reaction to the system.

It is an automated workflow that ensures dynamic and responsive restaurant sales for maximum profitability.

II. LITERATURE SURVEY

Literature work related to Weather and Restaurant Sales

The influence of weather on the behavior of customers has been described and shown by many authors in different contexts. For instance, Bujisic et al. [1] proved that the change in weather primarily affects the foot traffic into restaurants as more considerable pleasant weather motivates customers to go out for eating. Similarly, Rose and Dolega [2] analyzed a spatial analysis to quantify the impact of weather on retail sales that is represented by the demand of having a weather-based model towards forecasting requirements. Yang et al. [3] focus on incorporating weather data into sales prediction models by using advanced feature interaction learning mechanisms to include the impact of weather.

Some of the studies include work by Tanizaki et al. [4] and Schepers et al. [5], which explores how diner behavior may be predicted using weather patterns, pointing out that comfort foods sell more during colder weather, while lighter meals sell better during warmer weather. Nassibi et al. [6] argue that if restaurant chains include weather data in demand forecasting models, it will enable restaurants to understand their customers' preferences better, hence menu items are managed more appropriately.

Customer responsiveness is greatly enhanced in terms of response to dynamic changes in menu items with respect to the weather, as reviewed from Zháo and Jayadi's [7] demand forecasting for an Indonesian restaurant chain.

Literature work related to Ingredient Prices with Dynamic Pricing

One of the most significant challenges to restaurant profitability is dynamic ingredient prices. Singh et al. [8] expounded on a model for dynamic pricing strategy, tailored to Indian Railways, but generally applicable to restaurants. The authors' approach is underscored by the need for dynamic prices based on supply chain alterations. Popescu and Wu [9] worked on dynamic pricing with reference effects. This should work well in restaurant ingredient price fluctuations without antagonizing the customers.

Banerjee and Gurung [10] analyzed whether such models like ARIMA and VAR could predict the price of vegetables, which is one of the basic needs for restaurants, which have fresh produce as a source. Namasudra et al. [11], as well as Deksnyte and Lydeka [12], were also eager to make the fact that dynamic model pricing for the alteration in both supply chain and demand of consumers helps businesses incur profits without making prices burdening on the customer.

It extends these concepts by incorporating ingredient cost data coupled with the real-time sales and weather patterns that throw out a holistic pricing model. The project draws inspiration from Mishra et al. [13] as well as Yamuna et al. [14] and aims at the development of a dynamic framework that tends to adjust the prices based on the interaction between the external factors such as weather and the internal drivers such as ingredient costs.

Literature work related to Machine learning for forecasting sales

Machine learning has become an important tool to foresee sales and make restaurant operations more efficient. Pavlyshenko [15] studied different models of machine learning for time series predictions. It was shown to make them quite efficient in sales trend forecasting using only historical information. Schmidt et al. [16] follows the same line of using machine learning for restaurant sales forecasting. The study used neural networks for modeling, making it possible to handle complex nonlinear dependencies between various variables, such as weather, the prices of ingredients, and client preferences.

In fact, Tsoumakas [17] conducted a comprehensive review of the approaches machine learning applies in terms of predicting food sales, and indeed finds that ensemble methods such as Random Forests are extremely effective even at surpassing conventional statistical models in handling vast, multi-dimensional datasets. Rane et al. [18], among others, also outlined how ML may look to enhance customer satisfaction through the prediction of their behavior while adjusting pricing accordingly.

Other promising approaches include ARIMA models on time series forecasting, as suggested by Panda and Mohanty [19], and hybrid models, which combine machine learning techniques with statistical techniques, as suggested by Mahmoud and Mohammed [20].

Hence, these models can be exploited in the framework of Enhanced sales forecasting in order to enhance the predictions toward the sales during seasonal variations in customer demand.

III. METHODOLOGY

Methodology Steps:

1. Data Collection
2. Data Preprocessing
3. Model Selection
4. Model Training and Evaluation
5. Integration for Real-Time Analysis

Data Collection

First, the actual collection of detailed datasets that will be applied to the project will begin. Historical sales data will be used from the restaurant, showing the different categories such as highly perishable, perishable, non-perishable, and imported goods. Also, weather conditions, which involve temperature, humidity, and rainfall factors, will also be covered because external conditions play a big role in determining consumers' actions. This component shall collect ingredient price data and note the volatility from differences in circumstances or even the market drivers, potentially highlighting some supply-side-related challenges. The primary sources will come from real-time databases; governmental reports or publications will be used for good coverage of all factors. Here are the datasets employed for this project:

- chennai_1990_2022_Madras.csv Source: Kaggle

Historical weather data with climate parameters-temperature, about Chennai.

-FAOSTAT_data_en_10-05-2024*

FAOSTAT dataset with ingredients, producer price series: in this analysis of forces driving the prices.

Additional categorization done with the help of highly_perishable and non_perishable, perishable ,imported and price sensitivity.

These shall be combined for integrated analyses between internal sales and outside influences in the environment as preparation to take effective decisions.

Data Preprocessing

The preprocessing step is very intensive and begins post-gathering. In this process, cleaning data of all irregularities, or anomalies that are capable of deviating results, is followed by missing values being filled via either imputation or deletion. Here, features engineering take the lead whereby the proper attributes are defined or altered in such a way that they now do represent better relationships between variables that would be collected.

For instance, classification of products as high perishable, perishable, non-perishable, and imported can enable the model to understand how each type of product reacts in different circumstances. Then the datasets are brought together into a single integrated dataset with the sales records correlated to their respective weather and price records.

Model Selection:

Random Forest is selected considering strength for multiple input features with the feature importance which, in the case of this sensitivity and categorization analysis for the impact on sales, might be crucial. ARIMA was preferred as it is very much applicable to the time series model thus allowing to understand seasonal patterns in ascending order of time in sales. These algorithms together provide a comprehensive approach to understanding and predicting sales patterns.

Model Training and Evaluation

The next step is to train the selected models using the historical datasets. The data is typically split into training and testing subsets, allowing the models to learn from past sales patterns and evaluate their predictive accuracy. Hyperparameters are tuned for optimally maximizing performance, and the techniques of model validation used are cross-validation to ascertain that the model generalizes well to unseen data. Predictions' accuracy is checked by using key performance metrics, namely Mean Squared Error (MSE). Lower the MSE, the more efficiently the model is able to capture the underlying relationships between variables in the data.

Integration for Real-Time Analysis

Finally, the developed models are integrated into a system that enables real-time demand and dynamic pricing. Integration facilitates continuous processing of new inputs, including real-time changes in weather and changes in ingredient prices. Once more data becomes available, these models can adjust their forecast in a matter of seconds, making it possible for restaurant owners to take the right inventory or pricing decisions. This is especially important for perishable products, as timely adjustment can help in preventing waste and optimize sales. The system also has specific pricing strategies depending on the type of product. Imported items and price-sensitive products are well taken care of.

IV. EXPERIMENT AND ANALYSIS

IMPLEMENTATION OF DATA VISUALIZATION FOR INSIGHT EXTRACTION

Density Plot Description:
Comparing the actual and the fitted price distribution by the regression model. The actual prices are plotted in a filled KDE curve in blue colors, and the fitted ones in orange filled colors so they are differentiated by semi-transparent color fillings.

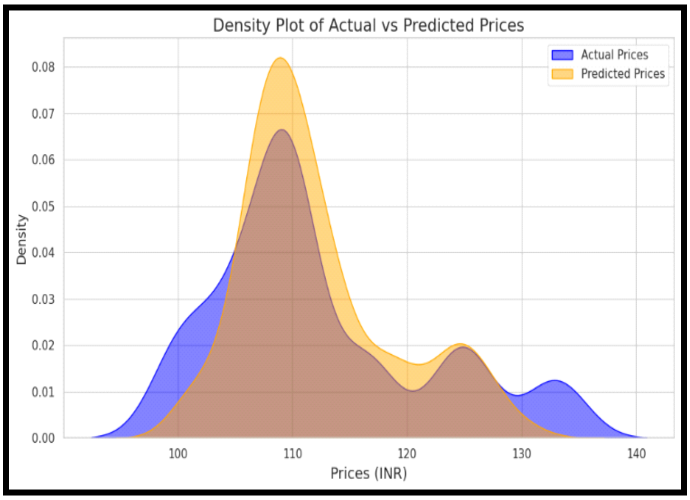

Figure 3 : Density Plot of Actual and Predicted Prices

Figure 3 depicts the closeness of actual prices to the values that were predicted as model performance. Similar distributions represent the rightness of the model, and large deviations are representative of error areas in the model.

Histogram Description:
Plot histogram of error distribution of the regression model. The frequency of different error values is shown with 30 bins, and an overlay of KDE will smooth the error distribution.

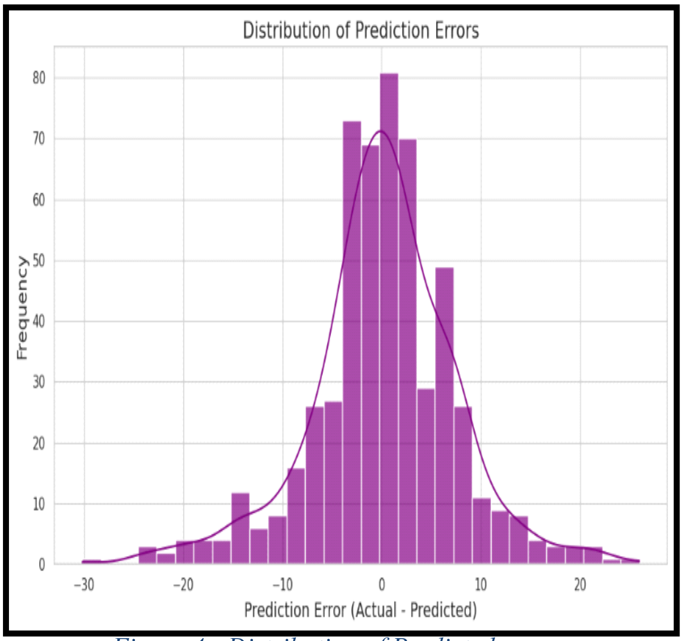

Figure 4 : Distribution of Predicted errors

Figure 4 depicts the extent to which model predictions can be accurate by bringing out both the frequency and dispersion of error. A distribution hugging closely around zero with relatively low dispersion would indicate higher accuracy while greater dispersion, or more significantly skewness suggests where adjustments in the models might need to be incorporated.

Bar Plot Description:
The above is a bar chart that represents the evaluation metrics for the Random Forest regression model, including MAE, MSE, and R² Score. All of them are in different colors so that it will be easier to compare.

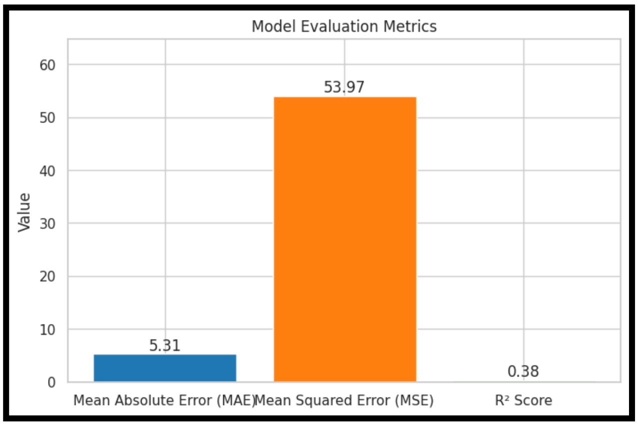

Figure 5 : Model Evaluation Metrics

Figure 5 shows a short visual summary of model performance. Lower MAE and MSE values relate to lesser errors, and a higher R² Score implies more effectiveness of the model in explaining variance. This evaluates the accuracy of the model and identifies where it needs improvement.

Heatmap Description:
A heat map of the price sensitivity level of various items. To the right is the color bar, which describes a range for the values on sensitivity.

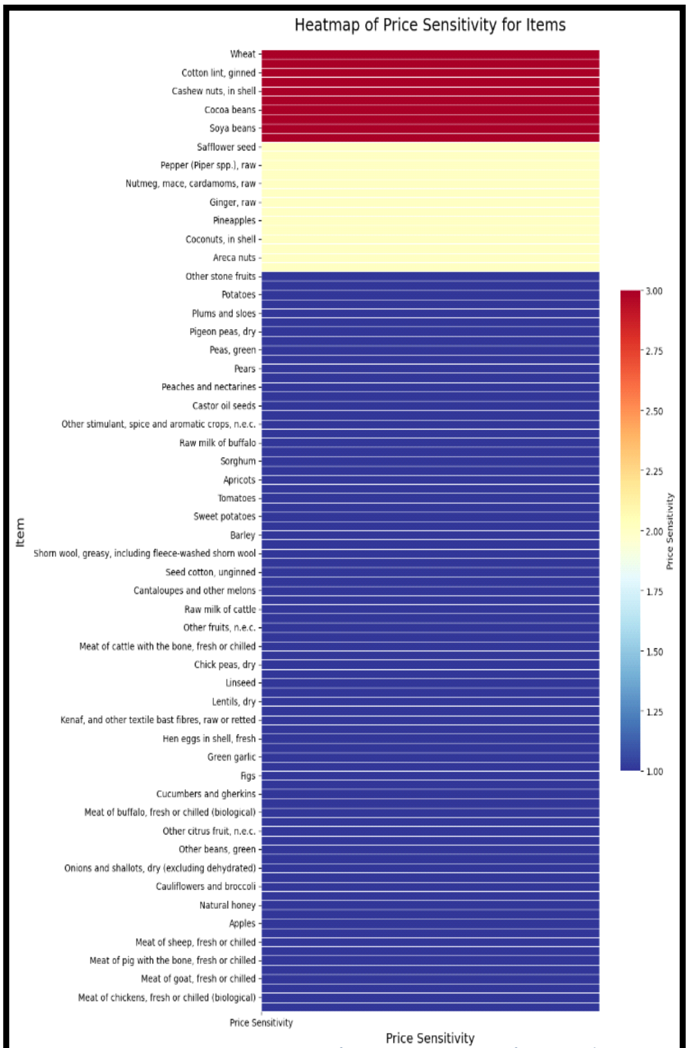

Figure 6 : Heatmap of Price Sensitivity for Goods

Figure 6 heatmap is very efficient to represent how food products differ from each other concerning the sensitivity to price change. The darker or lighter shade explains what items are relatively more or less sensitive in their response to a variation in prices, and that therefore, helps strategic decisions concerning price as well as market segmentation.

Description of Line Plot:
A line plot showing the trends of imports over time for various perishable items. Each line represents an item and the changing import quantity every year.

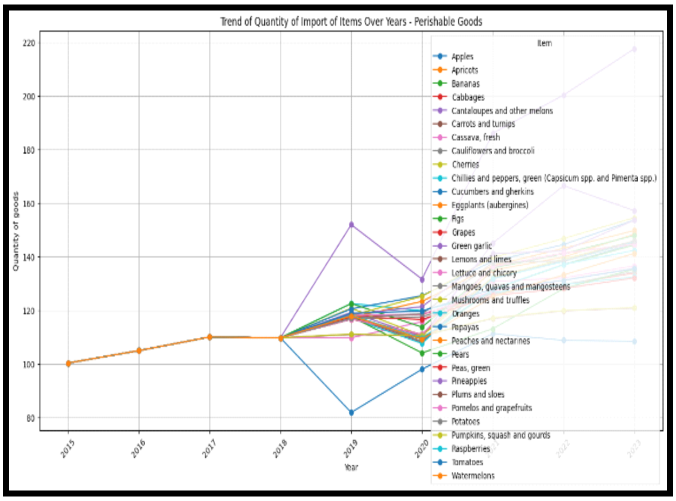

Figure 7 : Trends of Imported items over years on Perishable Goods

Figure 7 shows trends in volumes of different perishable items imported, and one can easily compare volumes across years. Fluctuations or steady trends point to insights that may influence supply chain decisions or change the import strategy.

Description of Trend Visualization:
A FacetGrid of line plots of the annual pattern of imported quantities for some very perishable items. One plot per product; extra data points to render all points visible.

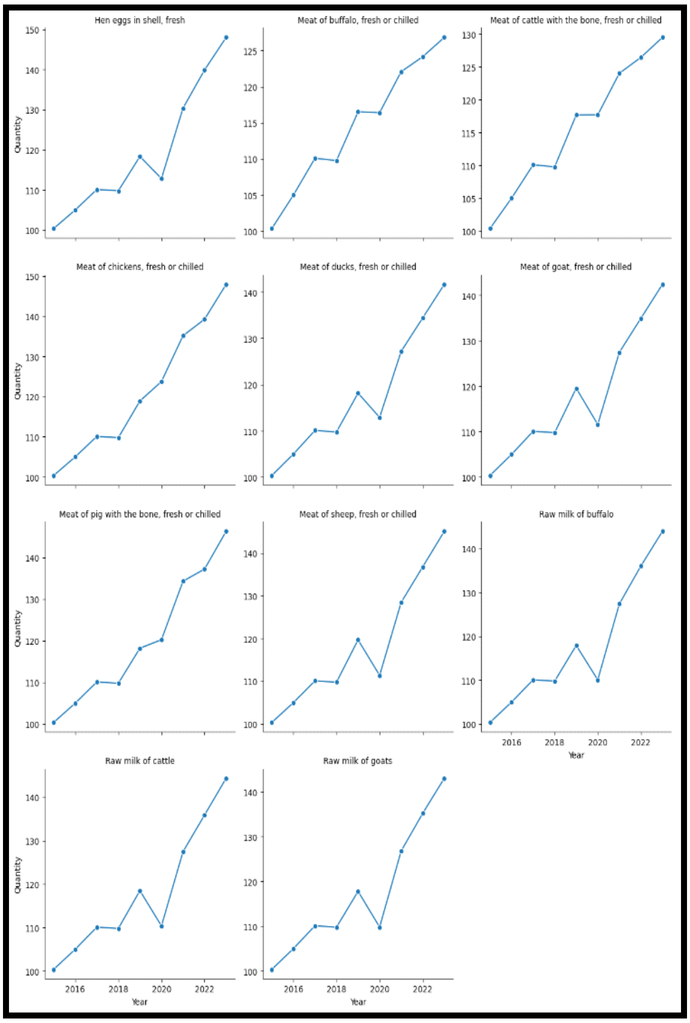

Figure 8 : FacetGrid Creation for Highly Perishable goods

Figure 8 line plot shows that for every product, the rate of importation is different. This plot makes it relatively easy to compare items through time. Steep climb or fall for some may be highly informative for inventory and import planning strategy.

Line Plot Explanation:
A line plot of the import trends of various non-perishable items over the years. Each item is represented by a line, and circular markers are given at each point for easy readability.

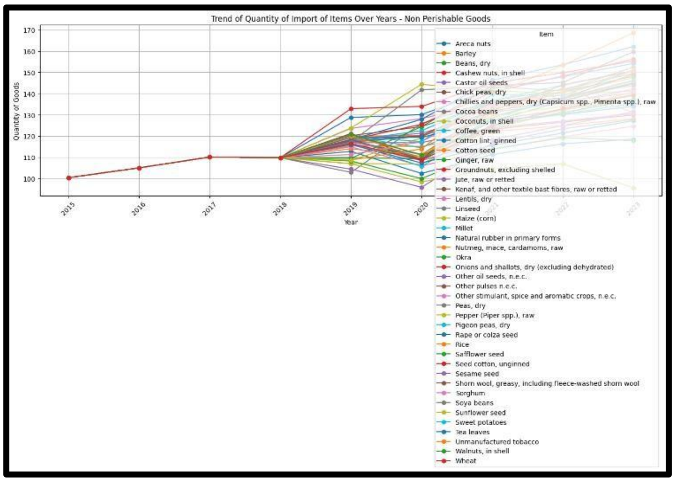

Figure 9 : Trends of Imported items over years on Non - Perishable Goods

The figure 9 depicts the import curves, peak, and collapse for the non-perishable goods, thus affording a trend analysis to be made on the general products. Major volume shift over time can help the organization make appropriate adjustments with the inventory and supply chain, thus optimizing the non-perishable stock's management.

The line graph shows the pattern of the import volumes of the different types of non-perishable goods over the years. Each line that is a different color represents a different item and is denoted by a circular symbol to allow for easier visibility on the time axis. The change of lines over time is, from 2018 on, and very clearly shows a divergence among the products - some with marked increases in import volume and others had no increment and even a decreased volume. It is apparent there are diverging demand and supply of these products that also could involve restrained domestic production by consumer trends, international market preferences, trade agreements, or geopolitics.

The parallels of the graph also show that the imports of non-perishable imports were more similar in earlier years since they clustered in terms of volume, but by later years there is more of a spread of lengths, which illustrates that there were more common items of imports in earlier years, but, over time, has become more specialized by product types. This further emphasizes the development of tailored forecasting for the specific products instead of a general idea of demand for a category of items.

The researchers looking at this will be able to see patterns in the trends over the years they can allow for planning to prepare for disruption, see opportunities for growth of products or other supply shortages or whether the stock of popular items has established ways to present to the agent, but without filling storage facilities with missing demand stock or incurring costs from supplying items that are not moving.

To further validate the work, the architecture developed was tested with few real time data collected from 10 different restaurants and our cafeteria, hostel mess.

V.RESULT AND DISCUSSIONS

TABLE 1:

Metric	Value
Mean Absolute Error (MAE)	5.31
Mean Squared Error (MSE)	53.97
R^2 Score	0.38

Table shows the performance metrics of the sales predictive model developed through Mean Squared Error, and R-squared
For any model, different metrics such as MSE or R2 provide different insights into a model.

MSE: This is an average squared difference between predictions and actual sales values. If the MSE is very low, it means the model is making predictions quite close to the observed value. The error is minimal. Results obtained in our study also reflected the fact that model values captured the trends related to the sales data, specifically those for perishable as well as highly perishable products, which are naturally sensitive to changes in such extraneous factors as weather.

R-Squared (R^2): This R^2 metric explains the variance in sales that could be contributed or explained by the model's inputs or inputs comprising its variables, such as weather variables and price sensitivity. The greater the value of R^2, the stronger the predictive strength of the model in capturing a factor that influences sales. From this analysis, it can be derived that R^2 varies category wise. Each category exhibits the sensitivity to the extent to which it reacts on variations in weather and change in price. For example, R^2 for more perishable products is rather higher, showing that there's a great aptness in the model for describing how the demand for it reacts to weather conditions changes.

Overall, results of this study indicate predictability to be the highest for highly perishable items, followed by perishable, and lowest in the case of non-perishable. This is intuitive with the expectation that the product like fresh fruits would have a lot of responses due to environmental factors like temperature and seasonality, and products such as canned goods would have fewer fluctuations. This confirms the concept that real-time pricing would help businesses maximise sales in sub-product lines sensitive to particular circumstances.

VI .CONCLUSION

This project provides a solid, data-driven solution to the restaurant industry because restaurants can immediately respond to changes in weather and the prices of ingredients. The model maximizes the management of perishable, non-perishable, and imported goods with the use of machine learning algorithms like Time Series Forecasting and Dynamic Pricing. Unlike static models applied earlier, this model will actually take real-time data, using predictive analytics to ensure the restaurants are responding well to the influence of extrinsic factors, but all of this without allowing it to compromise on the improvement in customer satisfaction, thus providing room for profit margins to be protected.

The restaurant managers, through the web of integrated interactive dashboards, establish consumer behavior and come up with fact-based decision-making about pricing as well as inventory levels. With the current weather dynamics, pricing and promotions will attract more customers, generating revenues. The entire project delivers a forward-looking approach, equipping restaurants to handle an ever-competitive market towards long term success and adaptability given the changing landscape of an industry.

VII .FUTURE ADVANCEMENT

The project will look to the future to improve prediction accuracy by using ensemble learning and deep neural networks to identify complex patterns in sales and weather data. Additionally, Power BI will be extended for interactive visual dashboards to give deeper insights in real-time data- driven decisions. These developments can be of great help to the restaurant industry in terms of optimizing ingredient inventory, dynamic pricing, and the sale strategy based on the actual demand forecast.

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